TigerGraph High-Availability - Node-Failure Test Report

A 3-node TigerGraph 4.1.4 cluster was deployed with high availability (replication factor 2, 1 partition / 2 replicas) and tested under node failures. 17 test cases cover functional behaviour, failure behaviour, and negative/boundary conditions; all pass. Each failure scenario was executed three times; medians are reported. Under HA, freeze, network-partition and single-component failures run at 100% availability with zero downtime (a replica serves), while losing a live node - including the master - costs a ~24 s median failover window (worst observed 44 s), versus ~52 s outages without HA.

HA vs non-HA (availability / MTTR, median of 3 runs)

Failure	Non-HA (RF=1)	HA (RF=2)
Freeze (pause)	23 s, 87%	0 s, 100%
Network partition	30 s, 86%	0 s, 100%
Component (GPE)	56 s, 86%	0 s, 100%
Data-node crash	52 s, 74%	24 s, 95%
Graceful stop	53 s, 74%	24 s, 95%
Master-node crash	53 s, 74%	24 s, 95%

Execution results (all 17 cases PASS)

ID	Type	Description	Status	MTTR	Avail	Notes
TC-QL-001	Positive	Point lookup: query a Person by primary id returns exactly one vertex.	PASS	-	-	Point lookup returned exactly one vertex.
TC-QL-002	Positive	1-hop traversal: friends of a Person are returned.	PASS	-	-	1-hop traversal returned the neighbour set.
TC-QL-003	Positive	Aggregation: count of all Person vertices equals 5000.	PASS	-	-	Aggregation returned the full Person count.
TC-QL-004	Positive	Multi-hop traversal: 2-hop friendship expansion returns a result set.	PASS	-	-	2-hop traversal returned a result set.
TC-SV-001	Positive	Service health: all critical services (GPE, GSE, RESTPP, GSQL) are Online.	PASS	-	-	All critical services (GPE/GSE/RESTPP/GSQL) Online.
TC-LJ-001	Positive	Loading job: load Person rows from a CSV file source; vertex count increases.	PASS	-	-	LOAD SUCCESSFUL; vertex count increased by the rows loaded.
TC-NF-001	Failure	Data-node hard crash: kill a node; measure availability and MTTR.	PASS	23.86	94.89%	Recovered via replica failover. Median of 3 runs (MTTR 23.53-30.5 s).
TC-NF-002	Failure	Graceful node shutdown: stop a node; measure recovery.	PASS	23.68	94.89%	Recovered via replica failover. Median of 3 runs (MTTR 23.67-23.95 s).
TC-NF-003	Failure	Frozen node: pause a node (up but unresponsive); measure recovery.	PASS	-	100.0%	Zero downtime - replica served throughout. Median of 3 runs.
TC-NF-004	Failure	Network partition: isolate a node; measure rejoin behaviour.	PASS	-	100.0%	Zero downtime - replica served throughout. Median of 3 runs.
TC-NF-005	Failure	Single-component failure: stop GPE on one node; measure behaviour.	PASS	-	100.0%	Zero downtime - replica served throughout. Median of 3 runs.
TC-NF-006	Failure	Master/gateway-node crash: kill tg1; observe from a surviving node.	PASS	23.91	94.89%	Recovered via replica failover. Median of 3 runs (MTTR 23.88-43.97 s).
TC-SV-002	Failure	Service crash on node loss: killing a node takes its GPE/GSE/RESTPP instances down; a replica stays Online.	PASS	-	-	Killed node's service instances went down; a replica stayed Online.
TC-WR-001	Failure	Write durability under crash: writes during a node crash are not lost.	PASS	-	92.5% writes	No acknowledged write lost (durable=True).
TC-WR-002	Failure	Write durability under partition: ambiguous writes; nothing acknowledged is lost.	PASS	-	85.0% writes	No acknowledged write lost (durable=True).
TC-NG-001	Negative	Invalid query is rejected gracefully and does not crash the cluster.	PASS	-	-	Invalid query rejected with an error; gateway kept serving.
TC-BD-001	Boundary	Two-node failure exceeds RF=2 tolerance; cluster must recover after nodes return.	PASS	-	-	Availability lost with 2/3 nodes down; cluster recovered after both returned.